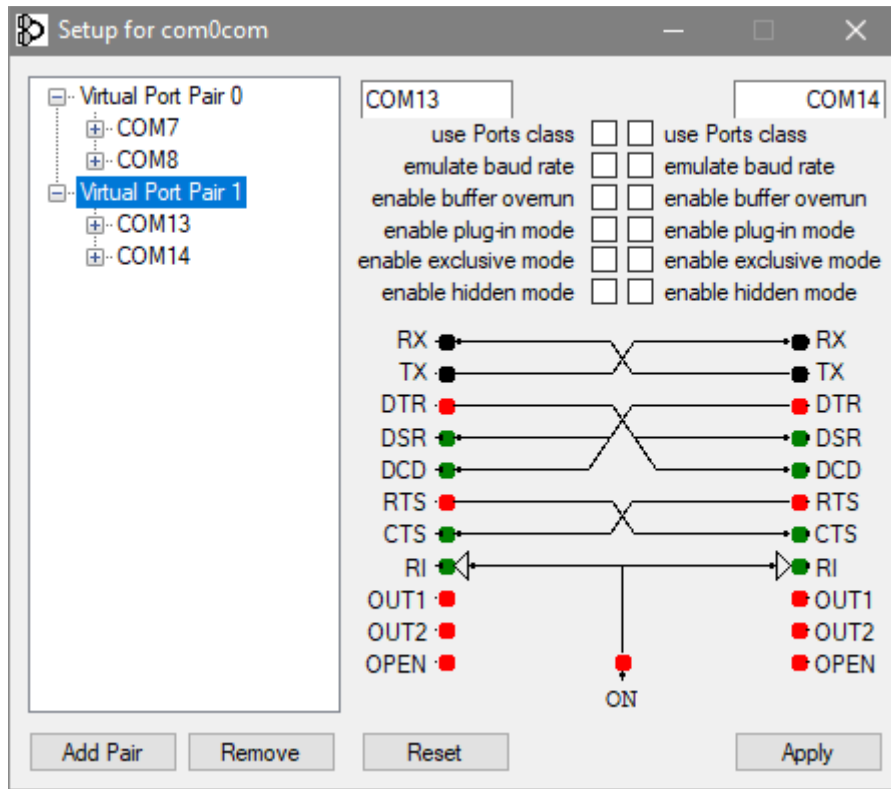


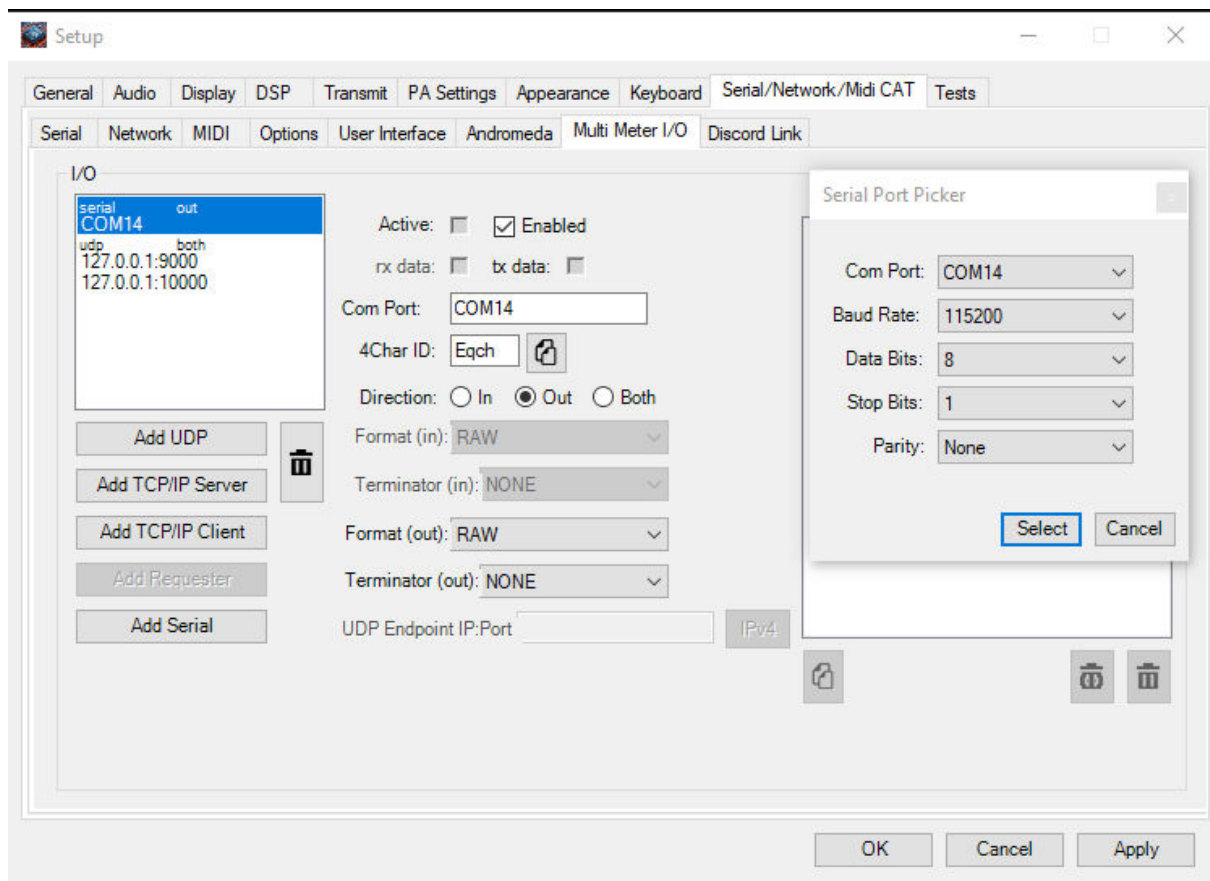
Phil's (VK7ZPD) Notes on Thetis and PA100D band change built on Luca's work (EA8DJR)

Control of the Juma PA100d is achieved using the server and client applications from Luca's initial instructions and using the same container.

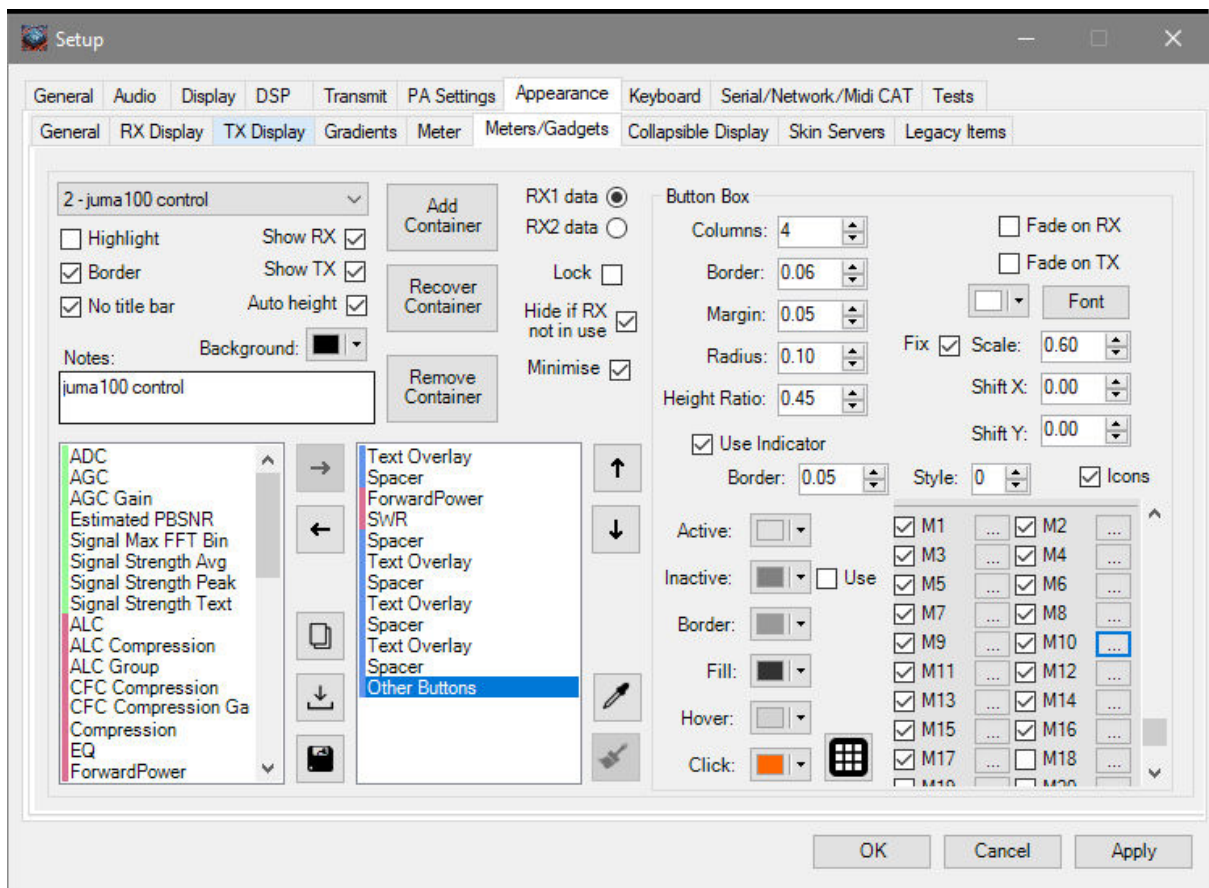
Set up a com0com virtual com port pair in this case I'm using com 13&14 for CAT control of the HL2



In the Multi Meter I/O section tab add one of the com0com ports com port (Add Serial) & make sure to note the port speed. Also note of the 4Char ID which will be needed in the macro buttons. Port speeds must match in I/O and Serial cat settings. You can check the port speed by clicking on the allocated com port.



Set up your macro buttons for band change. For clarity I've only shown only M10 the 80m button. If you check the no title bar option the container will be locked in place and the first field isn't blanked when hovering over that area.



Macro Button Configuration - Macro 10

ON state text: Notes:

OFF state text:

☐ Closes parent container
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size

Button State MMIO Actions CAT Macro

Button State

☐ On
☐ Off
☐ Toggle
☒ Led Indicator 4char
☐ Container visibility
☐ CAT (eg. use [STATE]PS; in macro)
 ON=
☐ Run state cmd when parent container becomes visible

OK Cancel

Set up the MMIO actions as follows. Note that the first message tells the PA100 to switch to 80m via the UDP server, the second tells the HL2 via cat to switch to 80m band. The cat commands are logical for example 40m is ;ZZBS040; etc The reason for adding the command in both the on and off is to ensure the command is sent on each press.

Macro Button Configuration - Macro 10

ON state text: 80m
OFF state text: 80m

Notes:

☐ Closes parent container
☐ Closes container 1 - Container
☐ Opens container 1 - Container
☐ use closed container for location + size

☐ Closes container 1 - Container
☐ Opens container 1 - Container
☐ use closed container for location + size

☐ Closes container 1 - Container
☐ Opens container 1 - Container
☐ use closed container for location + size

☐ Closes container 1 - Container
☐ Opens container 1 - Container
☐ use closed container for location + size

Button StateMMIO ActionsCAT Macro

MMIO Actions

	4char	ON raw text message	OFF raw text message
☒ Send message via	xiDM	=B2	=B2
☒ Send message via	Eqch	;ZZBS080;	;ZZBS080;
☐ Send message via			
☐ Send message via			

OK

Cancel

In the cat control Serial tab add the other half of the com0com port pair.

Also added are 3 80m Net frequencies, which change by the same CAT method as the bands. Note that the cat command frequency change string is 11 digits long, in this case the command for 3567 khz is ;ZZFA00003567000;

Macro Button Configuration - Macro 20

ON state text: Notes:

OFF state text:

☐ Closes parent container

☐ Closes container 1 - Container ☐ Opens container 1 - Container ☐ use closed container for location + size

☐ Closes container 1 - Container ☐ Opens container 1 - Container ☐ use closed container for location + size

☐ Closes container 1 - Container ☐ Opens container 1 - Container ☐ use closed container for location + size

☐ Closes container 1 - Container ☐ Opens container 1 - Container ☐ use closed container for location + size

Button State MMIO Actions CAT Macro

MMIO Actions

	4char	ON raw text message	OFF raw text message
☒ Send message via	xiDM	=B2	=B2
☒ Send message via	Eqch	:ZZFA00003640000;	:ZZFA00003640000;
☐ Send message via			
☐ Send message via			

Setup

General Audio Display DSP Transmit PA Settings Appearance Keyboard Serial/Network/Midi CAT Tests

Serial Network MIDI Options User Interface Andromeda Multi Meter I/O Discord Link

CAT1 Control ☐ VFO B

☐ Enable CAT1

Port: COM7 Baud: 115200 Parity: none Data: 8 Stop: 1

CAT2 Control

☒ Enable CAT2

Port: COM13 Baud: 115200 Parity: none Data: 8 Stop: 1

CAT3 Control

☐ Enable CAT3

Port: None Baud: 115200 Parity: none Data: 8 Stop: 1

CAT4 Control

☐ Enable CAT4

Port: None Baud: 115200 Parity: none Data: 8 Stop: 1

PTT Control

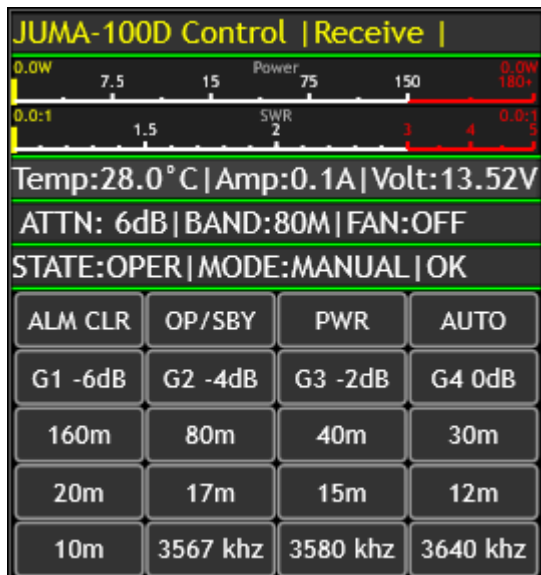
☐ Enable PTT

Port: None

☐ RTS ☐ DTR

OK Cancel Apply

The finished container will look something like this, I dock mine at the side of Thetis,



Clicking on any of the band change buttons should then change the HL2 band and the Juma.

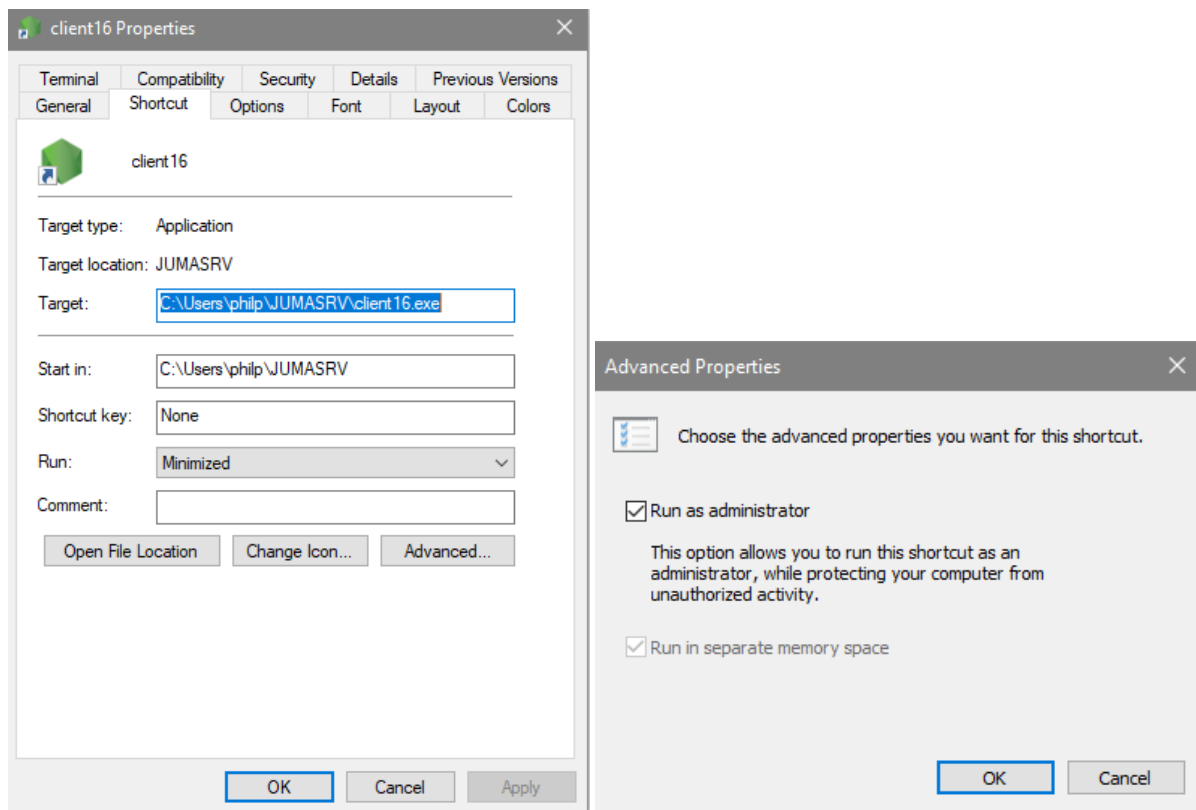
Some additional note. Rather than use the Thetis start options I activate the Server and client via shortcuts and batch files, the reason is that using that method allows those services to be activated as run as administrator and minimised, similarly I close those applications with batch file.

I also set up some container buttons for NR, Setup and Memory as this makes changes quicker than using the Setup menu. Having buttons for NR has an additional advantage that is toggle on and off the desired NR without needing to cycle through the NR options in Thetis.



Batch file start of the server and client are as follows.

Set up shortcuts to the server.batch and client files and flag as Run as Administrator and Run Minimised, the server and client will minimise to the taskbar. The reason for using shortcuts is that batch files can't be run as administrator.



Run both shortcuts with a batch file, in my example below I power on my PSU, run Voicemeeter for audio , and launch Thetis in the first commands.

The start commands launch the server and client with appropriate priviledges and minimised.

The batch file I use.

```
python c:/belfry/power500_switchingon.py
```

```
start c:\belfry\Voicemeeter.lnk
```

```
start c:\belfry\Thetis-HL2.lnk
```

```
start "" "C:\Users\philp\Desktop\Juma Server start\node server.lnk"
```

```
wait 2
```

```
start "" "C:\Users\philp\Desktop\Juma Server start\client16.lnk"
```


to shutdown the server and client I use the following.

```
taskkill /IM client16.exe /F
```

```
taskkill /IM node.exe /F
```

```
wait 10
```

Hope this helps.